

**Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination**

**INTRODUCTION TO MEMS**

Time : 3 Hours ]

[ Max. Marks : 60

**Instructions to Candidates :—**

- (1) Attempt all questions.
- (2) Assume suitable data wherever necessary.
- (3) Support your answer with suitable diagrams if needed.

1.     (a)     Distinguish between a micro sensor, an actuator and a system.             6(CO1)  
       (b)     What is miniaturization ? Why it is needed ?                             6(CO1)
  
2.     (a)     Suggest a suitable MEMS sensor for pressure sensing application; explain.             6(CO2)  
       (b)     For the purpose of depositing metals and other materials, illustrate and briefly describe the schematic of a thermal evaporation plant.             6(CO3)
  
3.     (a)     Discuss the difference between analog and digital RF MEMS capacitors.             6(CO4)  
       (b)     List the steps involved in MEMS fabrication using bulk micromachining, surface micromachining and LIGA process.             6(CO3)
  
4.     (a)     Describe the significance of the Hooke's law and the Poisson effect in solid mechanics for MEMS.             6(CO5)  
       (b)     What distinguishes essential boundary conditions from natural boundary conditions in Finite Element Method ?             6(CO5)

5. (a) What procedures are used to check for chemical content ? Of these, which one has been applied to microsystems ? Give appropriate justifications for your response. 6(CO2)
- (b) Elucidate RF MEMS components used in defense applications. 6(CO4)

